

Lab # 01
Getting Started with Linux Basics and Commands (Part B)



National University
of computer and emerging sciences

Operating Systems Lab (CL-2006)

Semester: Spring 2026

Section: BCS-4H

Course Instructor: Mr. Abdullah Shaikh

LAB # 01 – TASKS (B)

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Submission Instructions:

1. Perform all the assigned tasks carefully.
2. Prepare your solutions in a Microsoft Word document, clearly writing the solution of each problem.
3. Attach relevant screenshots for every task as proof of execution.
4. Ensure that your document is properly formatted (clear headings, proper spacing, and alignment).
5. After completing the Word document, convert it into a PDF file.
6. Upload the PDF file on Google Classroom (GCR) within the given deadline.
 - **No submission will be accepted after the deadline.**
7. Before uploading, rename your PDF file using the following format:
OS Lab 01 (B) – Your Name – Your Roll Number

Example:

OS Lab 01 (B) – Abdullah Shaikh – 25K-7826

TASK # 01

***Note:** For the following tasks, a separate text file is attached. Please use the provided text file to perform the tasks listed below.*

1. Find all occurrences of the word "linux" in the text, regardless of whether they appear as "Linux", "LINUX", or "linux". How many matches are there?
2. Display all lines from the text that do NOT contain the word "the" (case-insensitive). How many lines exclude this common word?
3. Count how many times the word "Python" appears in the text (case-sensitive).
4. Search for the exact word "go" in the text, ensuring you don't match words like "Google", "governance", or "going". How many exact matches exist?
5. Find all lines containing the word "security" and display their line numbers. Which lines contain this word?
6. Suppose the above text is split into three files: tech_part1.txt, tech_part2.txt, and tech_part3.txt. Search for "database" across all three files and show the matching lines.
7. Using the same three files from Question 6, search for "cloud" across all files but suppress the filenames in the output, showing only the matching content.
8. Find all lines that contain either "Java" OR "Python" OR "JavaScript". How many lines mention any of these programming languages?
9. Using the three files from Question 6, find lines containing either "server" or "database" across all files, showing which file each match comes from.
10. Using the three files from Question 6, display only the names of files that contain the word "encryption", without showing the matching lines themselves.
11. Create a file called keywords.txt containing three keywords: Linux, Python, and security. Now search the main text for any lines matching these patterns from keywords.txt.
12. Find all lines that begin with the word "The" (case-sensitive). How many lines start with this word?
13. Find all lines that end with the word "data" (case-insensitive). How many lines end with this word?
14. Imagine you have a directory called tech_articles/ containing 100 files with various extensions. Search recursively through all files in this directory to find occurrences of "artificial intelligence".
15. Use egrep (or grep -E) to find lines containing any of these terms: "AI", "Machine Learning", "Deep Learning", or "Neural Networks".

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16. (Bonus) Search for the word "blockchain" in the text, but suppress all output to the terminal (both matches and any error messages). Only use the exit status to determine if matches were found.

TASK # 02

Give a one-line description of each of the following process management commands in Linux.

1. ps
2. top
3. kill
4. nice
5. htop

TASK # 03

Write the cron job entry for the following scenarios.

(Write only the cron job command; no implementation required.)

1. Write a cron job to run a script named backup.sh every day at 2:00 AM.
2. Write a cron job to run a script named cleanup.sh every Sunday at 3:30 AM.
3. Write a cron job to run a script named monitor.sh every 10 minutes.
4. Write a cron job to run a script named log_cleanup.sh on the 1st and 15th of every month at 11:30 PM, but only if the day is not Sunday.

THE END!